

MONDIP SEN

Enterprise Cloud, Storage & Data Protection Solutions Engineer

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SUMMARY

Enterprise Cloud, Storage & Data Protection Solutions Engineer with **20+ years experience** designing, migrating, and operating enterprise backup, storage, and disaster recovery (DR) environments for Fortune 500 clients across healthcare, financial services, energy, consumer goods, and government. STS-certified on Veritas NetBackup, with hands-on depth across the full data-protection ecosystem — **NetBackup, Commvault, Veeam, Rubrik, Cohesity, IBM Spectrum Protect (TSM), Dell EMC NetWorker, Avamar, and PowerProtect DD** — protecting Linux, UNIX, Windows, VMware, and **AWS / Azure / GCP / OCI** workloads. Former **global data-protection team lead**, now a **Solutions Architect** delivering cloud-native, automated, observability-driven infrastructure with **Kubernetes, Terraform, Ansible, and CI/CD**.

CORE TECHNICAL SKILLS

- **Cloud Platforms & Containers:** AWS, Microsoft Azure, Google Cloud (GCP), Oracle Cloud Infrastructure (OCI); managed Kubernetes — **Amazon EKS, Azure AKS, Google GKE, Red Hat OpenShift, Helm, Docker**; AWS Backup, Azure Backup / Site Recovery; EC2/S3/EBS, MinIO; cloud HA and failover architecture
- **Infrastructure as Code & Automation:** Terraform, Ansible, Chef, Puppet; **GitOps** — Git as source of truth; **GitLab CI, Jenkins, GitHub Actions**; Python, Bash/KSH, PowerShell, SQL; Prefect workflow orchestration; systemd service engineering
- **Observability, SRE & Security:** Datadog, Splunk, IBM Instana, Prometheus, Grafana, Loki, Tempo, **OpenTelemetry**, Alertmanager; SLO and alert design; **HashiCorp Vault** (AppRole, envconsul); **Zero Trust** access, TLS/PKI; **HIPAA, PCI DSS, SEC 17a-4 / WORM retention, NIST & CIS Benchmarks**
- **Enterprise Backup & Recovery:** Veritas NetBackup 6.x–10.x (STS-certified), NetBackup Appliances (5230/5330), IBM Spectrum Protect (TSM), Dell EMC NetWorker, Avamar, Data Domain / **PowerProtect DD**, Veeam Backup & Replication, Commvault, **Rubrik, Cohesity**, NetApp AltaVault, Veritas Backup Exec
- **Storage, SAN/NAS & Virtualization:** Dell EMC VMAX / **PowerMax** / VNX / Unity, NetApp FAS / **ONTAP**, Pure Storage, Hitachi, HP, **TrueNAS, Unraid**; Brocade & Cisco Nexus SAN fabrics; LUN provisioning, multipathing, NFS/CIFS/iSCSI, Fibre Channel, ZFS; VMware vSphere 7/8, ESXi, vCenter, Hyper-V, **Proxmox VE, OpenStack**
- **Operating Systems & Databases:** Red Hat Enterprise Linux (RHEL), SUSE, Ubuntu, IBM AIX, HP-UX, Solaris, Windows Server 2008–2022; Oracle (RMAN), Microsoft SQL Server, Exchange, SharePoint, SAP, DB2, Sybase, Active Directory
- **Process & Tooling:** Agile / Scrum, Jira, Confluence, ITIL, ServiceNow, BMC Remedy, Zendesk; vendor & project management, service operational procedures

PROFESSIONAL EXPERIENCE

Enterprise Cloud, Storage & Data Protection Solutions Architecture — Bell *New York, NY (Remote) · Dec 2024 – Present* Integration

Solutions architecture across enterprise infrastructure, cloud, and data protection for clients including **Chemours, American Standard, and LIXIL**. Lead architect on **MTS1B**, a production AI and automation platform.

- **Architected and operate an enterprise zero-trust microservices platform on Kubernetes** spanning 16 containerized services, HashiCorp Vault secrets management across 18 paths, and end-to-end TLS encryption achieving **99.99% operational uptime**.
- **Engineered an enterprise observability & SRE framework:** Prometheus metrics, Grafana dashboards, Loki log aggregation, Tempo distributed tracing, and OpenTelemetry instrumentation, establishing SLO-based alerting and automated remediation runbooks.
- **Automated multi-cloud infrastructure provisioning** across AWS, Azure, GCP, and OCI using Terraform and Ansible with GitLab CI and Jenkins pipelines; orchestrated **90+ Prefect workflow deployments**, eliminating **85% of manual deployment toil**.
- **Enforced strict data integrity and governance** across multi-cloud workloads under a daily-sealed audit log meeting **WORM and SEC 17a-4 regulatory standards**, with **686 automated contract tests passing at 100%**.
- **Developed agentic AI / LLM automation pipelines in Python** featuring multi-provider routing, local retrieval-augmented generation (RAG), and prompt-cache cost engineering (FinOps) integrated with Datadog and Splunk.

Global Data Protection Team Lead / Sr. Data Protection Engineer — IPsoft (Amelia US *New York, NY · Mar 2016 – Nov 2024* LLC)

Enterprise backup and DR for Fortune 500 clients — **Estée Lauder, McKesson, Moody's, Celesio EU, Duracell, BP, and Sands Resorts** — across healthcare, financial services, energy, consumer goods, US federal, and cloud-provider sectors.

- **Directed global data protection engineering and DR operations** for Fortune 500 accounts (Estée Lauder, McKesson, Moody's, BP, Sands Resorts), managing cross-functional delivery in Agile/Scrum, engineer onboarding, and executive SLA reporting.
- **Architected, modernized, and operated 15+ PB enterprise backup estates** across Veritas NetBackup (6.x–10.x), Veeam Enterprise, IBM Spectrum Protect, Dell EMC PowerProtect / Data Domain, and NetApp AltaVault, safeguarding 10,000+ critical

SQL, Oracle RMAN, Exchange, and RHEL/Windows nodes.

- **Led enterprise business-continuity and DR failover audits**, consistently validating **RTOs under 45 minutes and RPOs under 15 minutes** through multi-site disaster simulation drills on NetBackup and Dell EMC Data Domain.
- **Resolved critical L3 storage and backup escalations** spanning NDMP, Oracle RMAN, Exchange, Sybase, and SQL Server; performance-tuned storage lifecycle policies and global deduplication to achieve **28% reduction in storage overhead**.
- **Administered multi-vendor SAN/NAS storage arrays** (Dell EMC PowerMax/Unity, NetApp ONTAP, Pure Storage, HP, Hitachi), orchestrating LUN provisioning, multipathing, and non-disruptive capacity expansions.

Sr. NetBackup Consultant — NYC Department of IT & Telecommunications (DoITT) *New York, NY · Sep 2015 – Feb 2016*

- **Spearheaded municipal disaster recovery deployment for NYC Government (DoITT)**, engineering clustered Veritas NetBackup 5230 appliances across NYC primary data centers and NJ DR sites protecting mission-critical municipal databases.
- **Automated backup validation and health-check workflows**, cutting daily backup failure rates from 14% to **under 0.5%** and ensuring 100% municipal audit compliance.

Sr. NetBackup Engineer / Enterprise Technology Consultant — Arrosoft Solutions *New York, NY · Feb 2013 – Jan 2016*

Data-protection deployment engineering for **Children's National Medical Center, NYC DoITT, FISA, NJ State Department of Health, Boston EHS, and Yeshiva University / Albert Einstein College of Medicine**.

- **Engineered multi-site enterprise storage and data protection environments** for Children's National Medical Center, NYC DoITT, FISA, NJ State DOH, and Yeshiva University, deploying NetBackup appliances, Auto Image Replication (AIR), and SAN storage networks.
- **Implemented EMC VMAX SRDF active-passive replication** with Veritas Cluster Server (VCS), validating seamless application and storage failover across primary and DR data centers for Children's National Medical Center.
- **Automated hyperscale server provisioning as DevOps Automation Engineer** using Chef, Puppet, PowerShell, and KSH; built cloud HA failover across AWS EC2/S3, Azure, and VMware with centralized syslog and network IDS.

Earlier Technical Roles

2009 – 2014

- **Hyper-V / Citrix VDI Support Engineer**, Diligex Cloud Services (clients: Viacom, US Space) — managed-service cloud VDI on Hyper-V and Citrix with AWS EC2/S3.
- **Network & Security Engineer**, GetFugu — scalable cloud-rendering infrastructure, intrusion detection, penetration testing.
- **Windows Network Support Analyst**, Star Alliance Capital (40 Wall Street) — trading-floor support for **200+ NYSE high-frequency equity and options traders** (Lightspeed/DAS).
- **Linux/UNIX System Administrator**, Intersessions Hosting — UNIX and Solaris hosting administration.

CERTIFICATIONS & TRAINING

- **Veritas / Symantec:** NetBackup for UNIX, STS Certified (2014) · Symantec Certified Specialist
- **Veeam Certified:** VMSP and VMTSP (2015) | **VMware:** VCP6 (vSphere)
- **NVIDIA Certified (2026):** AI Factory Deployment Learning Path · Cumulus Linux Essentials · SONiC Essentials · Networking Technical Curriculum
- **Automation & Systems:** IPsoft IPCenter CSI Automation (DevOps) & ICA | Arrosoft Certified Engineer: Backup, Storage & Virtualization

EDUCATION

B.Tech, Computer Information Systems — CUNY, New York City College of Technology, New York, NY (2010)

OPEN SOURCE & PORTFOLIO

- **github.com/senmondip** — publishing an open-source portfolio of enterprise data-protection and infrastructure automation: **38 repositories covering NetBackup, Commvault, IBM Spectrum Protect, Veeam, VMware vSphere, Linux/Windows Server, AWS, storage operations, disaster recovery, and Prometheus/Grafana observability**, released on a scheduled cadence through 2026.
- **Contributor — MH-VTL GUI** (virtual tape library): re-engineered the codebase for release builds.